

CLASS ACTIVITY 1

Scenario 1 – Student Academic Performance Dashboard:

Program:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

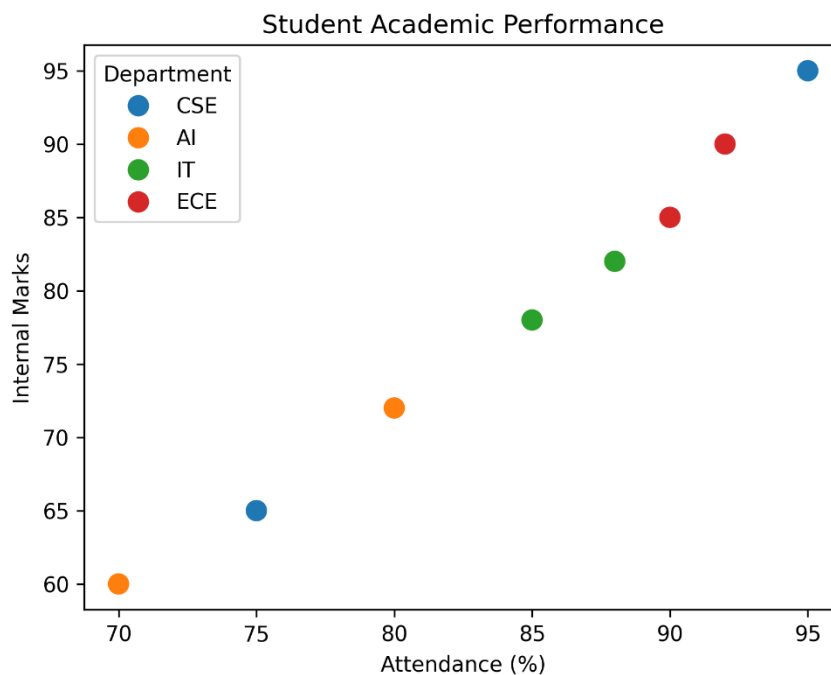
data = pd.DataFrame({
    'Attendance':[75,80,85,90,95,70,88,92],
    'Marks':[65,72,78,85,95,60,82,90],
    'Department':['CSE','AI','IT','ECE','CSE','AI','IT','ECE']
})

sns.scatterplot(data=data,
                x='Attendance',
                y='Marks',
                hue='Department',
                s=120)

plt.title("Student Academic Performance")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")

plt.savefig("scenario1.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:



Scenario 2 – Electric Vehicle Market Analysis:

Program:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

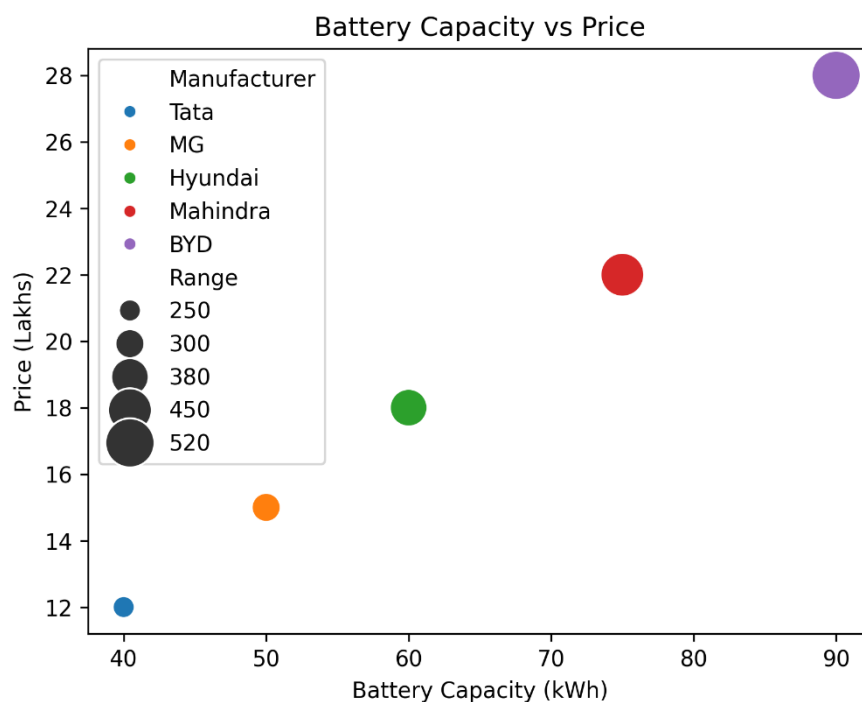
data = pd.DataFrame({
    'Battery':[40,50,60,75,90],
    'Price':[12,15,18,22,28],
    'Range':[250,300,380,450,520],
    'Manufacturer':['Tata','MG','Hyundai','Mahindra','BYD']
})

sns.scatterplot(data=data,
                x='Battery',
                y='Price',
                hue='Manufacturer',
                size='Range',
                sizes=(100,500))

plt.title("Battery Capacity vs Price")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price (Lakhs)")

plt.savefig("scenario2.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:



Scenario 3 – Hospital Patient Admission Analysis

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.DataFrame({
    'Department':['Cardiology','Neurology','Orthopedics','Pediatrics'],
    'Patients':[120,90,140,110],
    'Cost':[45000,52000,38000,30000]
})

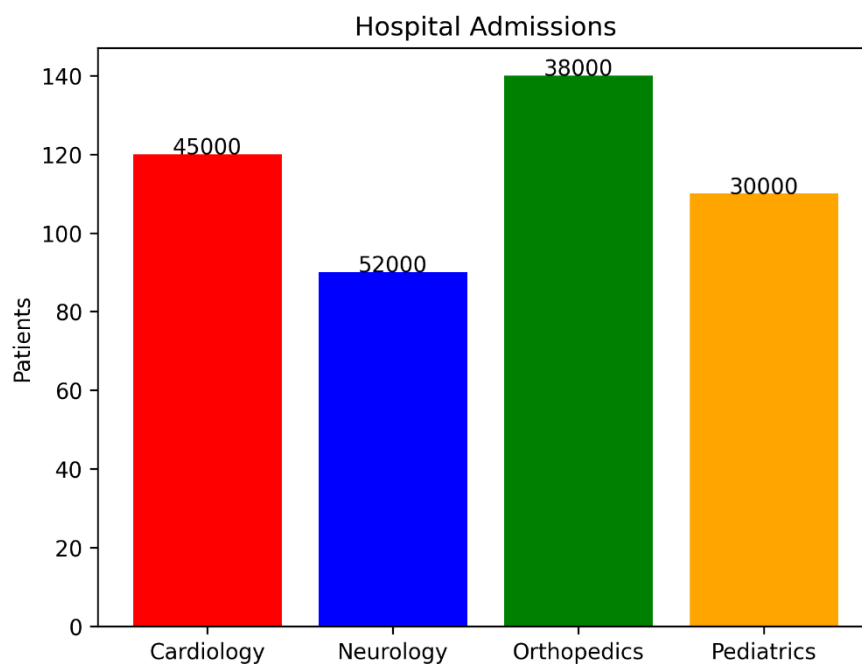
bars = plt.bar(data['Department'],
               data['Patients'],
               color=['red','blue','green','orange'])

for bar, cost in zip(bars, data['Cost']):
    plt.text(bar.get_x()+bar.get_width()/2,
             bar.get_height(),
             str(cost),
             ha='center')

plt.title("Hospital Admissions")
plt.ylabel("Patients")

plt.savefig("scenario3.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:



Scenario 4 – Weather Monitoring System:

Program:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

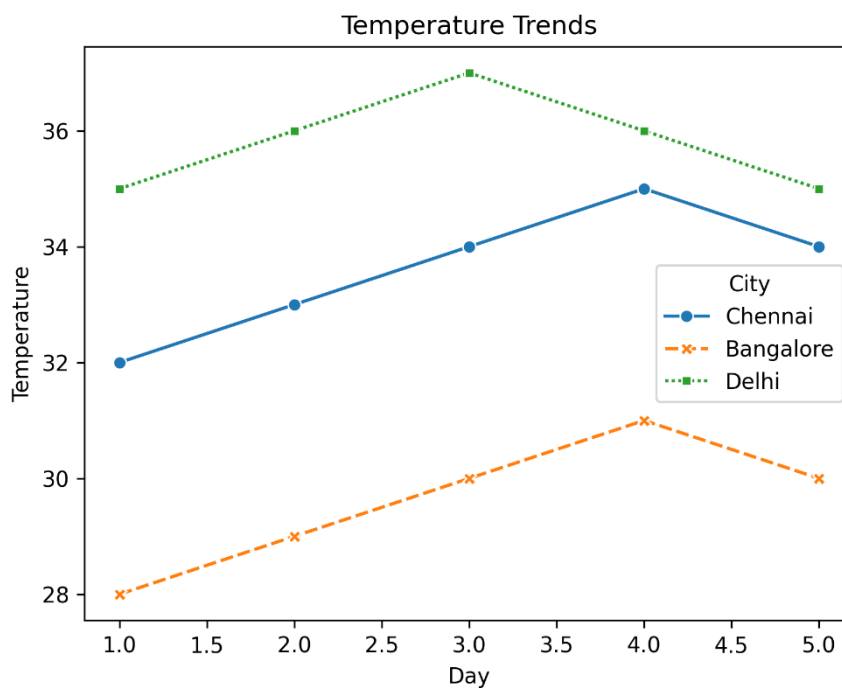
data = pd.DataFrame({
    'Day': [1,2,3,4,5]*3,
    'Temperature': [32,33,34,35,34,
                    28,29,30,31,30,
                    35,36,37,36,35],
    'City': ['Chennai']*5 +
            ['Bangalore']*5 +
            ['Delhi']*5
})

sns.lineplot(data=data,
              x='Day',
              y='Temperature',
              hue='City',
              style='City',
              markers=True)

plt.title("Temperature Trends")

plt.savefig("scenario4.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:



Scenario 5 – Supermarket Sales Analysis

Program:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

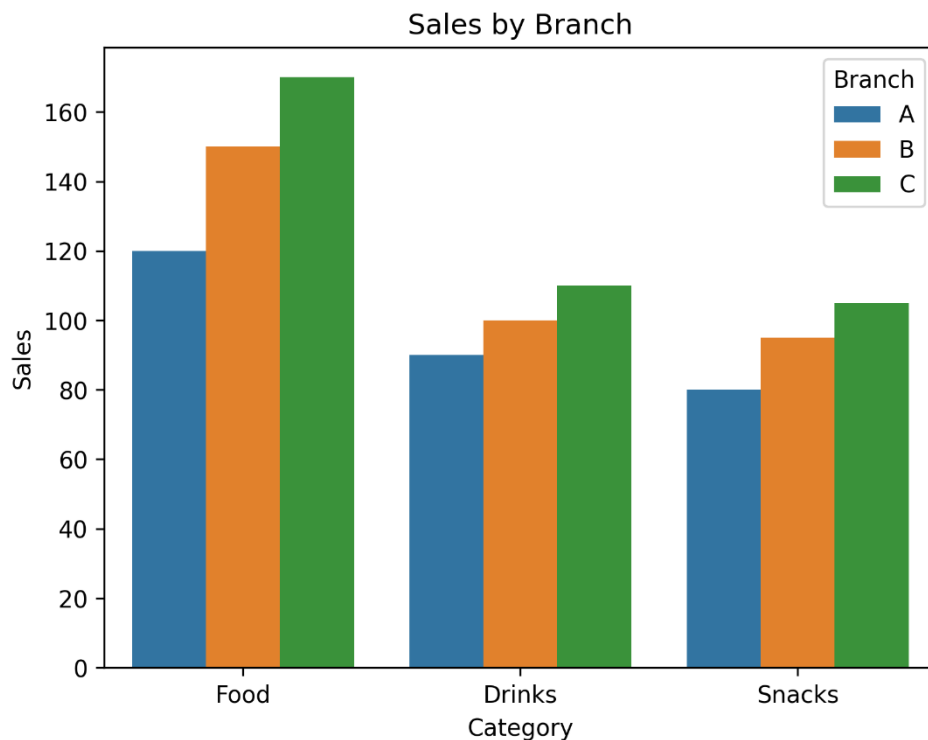
data = pd.DataFrame({
    'Category': ['Food', 'Food', 'Food',
                'Drinks', 'Drinks', 'Drinks',
                'Snacks', 'Snacks', 'Snacks'],
    'Sales': [120, 150, 170, 90, 100, 110, 80, 95, 105],
    'Branch': ['A', 'B', 'C'] * 3
})

sns.barplot(data=data,
            x='Category',
            y='Sales',
            hue='Branch')

plt.title("Sales by Branch")

plt.savefig("scenario5.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:



Scenario 6 – Employee Salary Distribution

Program:

```
import numpy as np
import matplotlib.pyplot as plt

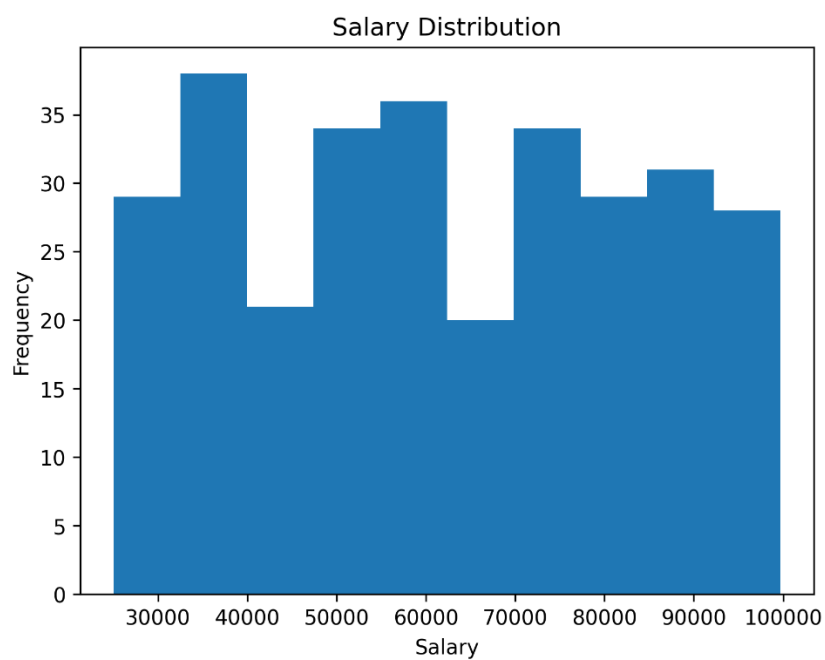
salary = np.random.randint(25000,100000,300)

plt.hist(salary, bins=10)

plt.title("Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Frequency")

plt.savefig("scenario6.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:



Scenario 7 – Smartphone Market Comparison

Program:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

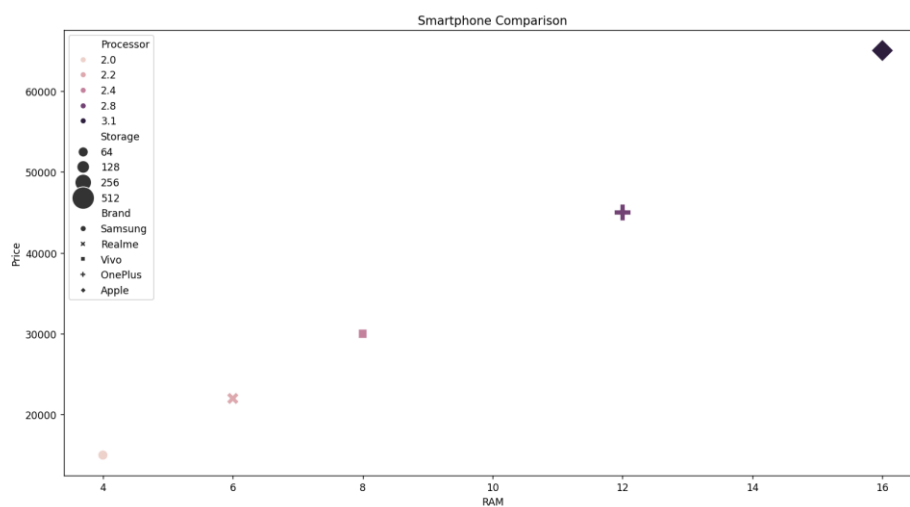
data = pd.DataFrame({
    'RAM':[4,6,8,12,16],
    'Price':[15000,22000,30000,45000,65000],
    'Storage':[64,128,128,256,512],
    'Processor':[2.0,2.2,2.4,2.8,3.1],
    'Brand':['Samsung','Realme','Vivo','OnePlus','Apple']
})

sns.scatterplot(data=data,
                x='RAM',
                y='Price',
                hue='Processor',
                size='Storage',
                style='Brand',
                sizes=(100,500))

plt.title("Smartphone Comparison")

plt.savefig("scenario7.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:

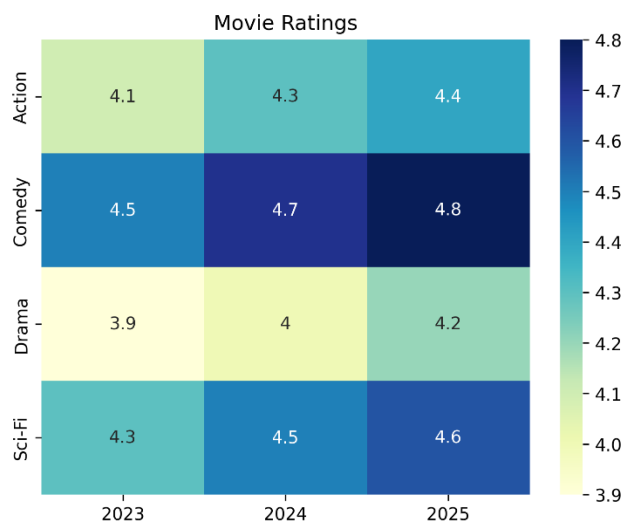


Scenario 8 – Online Movie Rating Analysis

Program:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
ratings = pd.DataFrame({
'2023':[4.1,4.5,3.9,4.3],
'2024':[4.3,4.7,4.0,4.5],
'2025':[4.4,4.8,4.2,4.6]
},
index=['Action','Comedy','Drama','Sci-Fi'])
sns.heatmap(ratings,
            annot=True,
            cmap='YlGnBu')
plt.title("Movie Ratings")
plt.savefig("scenario8.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:



Scenario 9 – COVID-19 State-wise Analysis

Program:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.DataFrame({
'State':['TN','KA','KL','AP','TS'],
'Cases':[1200,1800,900,1500,1100]
})

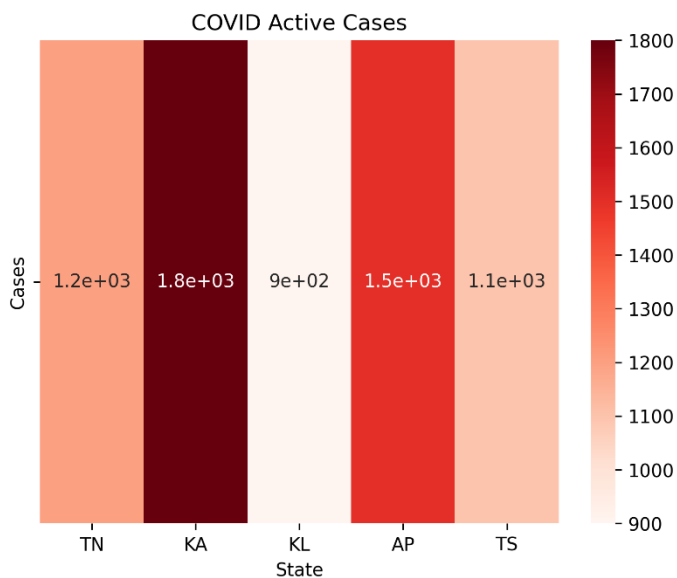
pivot = data.set_index('State').T

sns.heatmap(pivot,
            annot=True,
            cmap='Reds')

plt.title("COVID Active Cases")

plt.savefig("scenario9.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:



Scenario 10 – Iris Flower Classification

Program:

```
import seaborn as sns
import matplotlib.pyplot as plt
iris = sns.load_dataset("iris")
sns.scatterplot(data=iris,
                x='sepal_length',
                y='petal_length',
                hue='species',
                style='species',
                size='sepal_width',
                sizes=(30,250))

plt.title("Iris Flower Classification")
plt.savefig("scenario10.png", dpi=300, bbox_inches="tight")
plt.show()
```

Output:

